

ECSE-4040

Digital Electronics

Lecture 4

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Lecture Outline

- MOSFET I-V Characteristics
 - Short Channel
- MOSFET Capacitances
 - Gate Capacitances
 - Junction Capacitances
 - Overlap Capacitances
- DSM MOS IC Fabrication
 - Frontend (FEOL)
 - Backend (BEOL)

Electric Fields in Short-Channel MOSFET

1980	1995	2001
$E_y = \frac{5V}{5\mu m} = 10^4 V/cm$	$E_y = \frac{3.3V}{0.35\mu m} = 9.4 \times 10^4 V/cm$	$E_y = \frac{1.2V}{0.1\mu m} = 1.2 \times 10^5 V/cm$

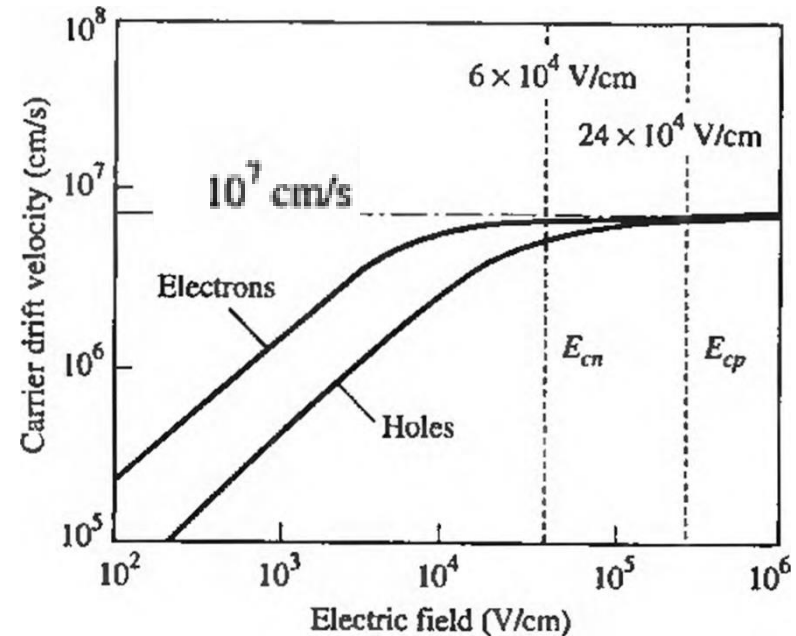
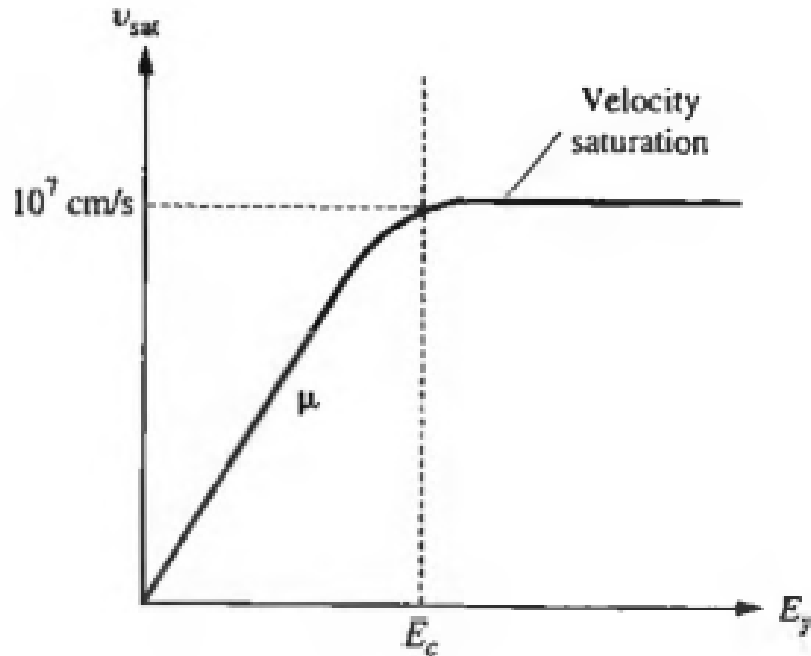
Horizontal Electric field (between the source and drain) = $V_{DS} (=V_{dd}) / L$

1980	1995	2001
$E_x = \frac{5V}{1000 \text{ \AA}} = 50 \times 10^4 V/cm$	$E_x = \frac{3.3V}{75 \text{ \AA}} = 4.4 \times 10^6 V/cm$	$E_x = \frac{1.2V}{22 \text{ \AA}} = 5.5 \times 10^6 V/cm$

Vertical Electric field (between the gate and channel) = $V_{GS} (=V_{dd}) / t_{ox}$

- Large electric field causes early carrier velocity saturation in channel of short-channel MOSFETs

Carrier Velocity Saturation



- Carriers reach limiting or saturation velocity (10^7 cm/s in Si) at high electric field
- Electrons saturate at a lower electric field than holes

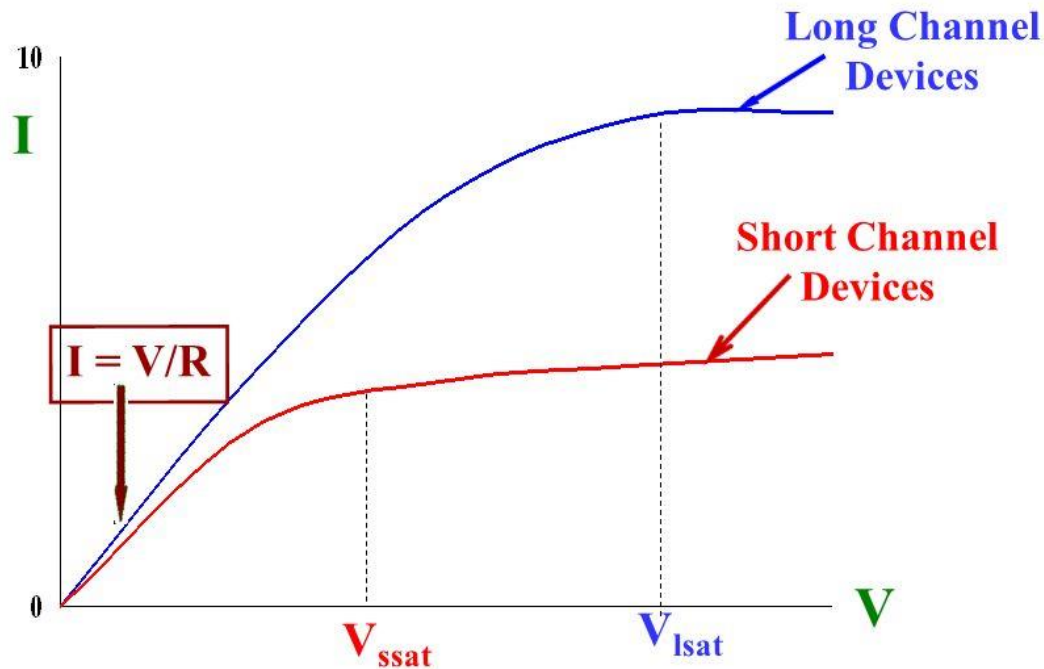
$$v = \mu_e \frac{E_y}{\left(1 + \frac{E_y}{E_c}\right)} \quad E_y < E_c$$

$$v = v_{sat} \quad E_y \geq E_c$$

$E_{cn} = 6 \times 10^4 \frac{\text{V}}{\text{cm}}$ for electrons
 $E_{cp} = 24 \times 10^4 \frac{\text{V}}{\text{cm}}$ for holes

MOSFET Saturation Current

Qualitative I-V Curves in Velocity Saturation Conditions



Current in Short-Channel MOSFETs

$$I_{DS} = W \times Q_n \times v \quad \text{General current of short channel MOS}$$

$$= W \times C_{ox}(V_{GS} - V_T - V(y)) \left(\frac{\mu_e E_y}{1 + \frac{E_y}{E_c}} \right) \quad \text{where } E_y = \frac{dV(y)}{dy}$$

Plugging in and re-arranging produces

$$I_{DS} dy = W \mu_e \left[C_{ox}(V_{GS} - V_T - V(y)) - \frac{I_{DS}}{W \mu_e E_c} \right] dV(y)$$

After integration, we obtain

$$I_{DS} = \frac{W}{L} \frac{\mu_e C_{ox}}{\left(1 + \frac{V_{DS}}{E_c L} \right)} \left(V_{GS} - V_T - \frac{V_{DS}}{2} \right) V_{DS}$$

- Similar to that for long-channel MOSFET except for extra term in the denominator

Current in Short-Channel MOSFETs

$$I_{DS} = W \times Q_n \times v_{sat}$$

Current of short channel MOS in saturation

Since the current is the same throughout the channel we can set $V(y) = V_{DS}$ and write

$$I_{DS} = W \times C_{ox} (V_{GS} - V_T - V_{DS}) v_{sat}$$

Equating this current and that of previous slide gives the required V_{DS} for saturation

$$V_{Dsat} = \frac{(V_{GS} - V_T) E_c L}{(V_{GS} - V_T) + E_c L}$$



Always smaller than $V_{GS} - V_T$
indicates early saturation

H&J Ex. 2.6

NMOS and PMOS Saturation Voltages for 0.18 μm Technology

Problem:

Consider a 0.18 μm technology. Compute the values of V_{Dsat} for the NMOS and PMOS device assuming $V_{GS} = 1.8\text{ V}$, $V_{TN} = 0.5\text{ V}$, $V_{TP} = -0.5\text{ V}$. Assume the channel length is 200 nm for convenience.

Solution:

Using (2.22), we find that $E_{cn}L_n = 6 \times 10^4 (0.2\text{ }\mu\text{m}) \approx 1.2\text{ V}$ and $E_{cp}L_p = 24 \times 10^4 (0.2\text{ }\mu\text{m}) \approx 4.8\text{ V}$. Using (2.28),

$$\text{NMOS:} \quad V_{Dsat} = \frac{(1.8 - 0.5)(1.2)}{(1.8 - 0.5 + 1.2)} \approx 0.6\text{ V}$$

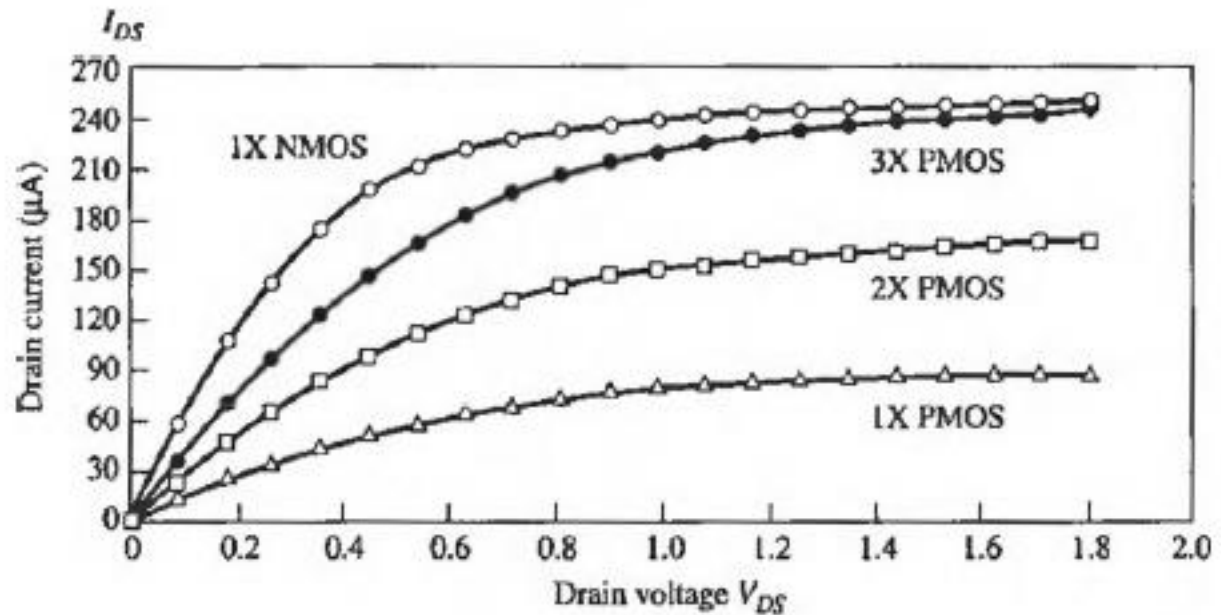
$$\text{PMOS:} \quad V_{Dsat} = \frac{(1.8 - 0.5)(4.8)}{(1.8 - 0.5 + 4.8)} \approx 1.0\text{ V}$$

- NMOS saturates faster than PMOS (at V_{DS} of 0.6 vs. 1.0V)

Current in Short-Channel MOSFET

- Electron channel mobility is about 4X higher than hole channel mobility, i.e., long channel NMOS current is 4X higher than PMOS of same size.
- However, early velocity saturation of electrons makes this ratio smaller.

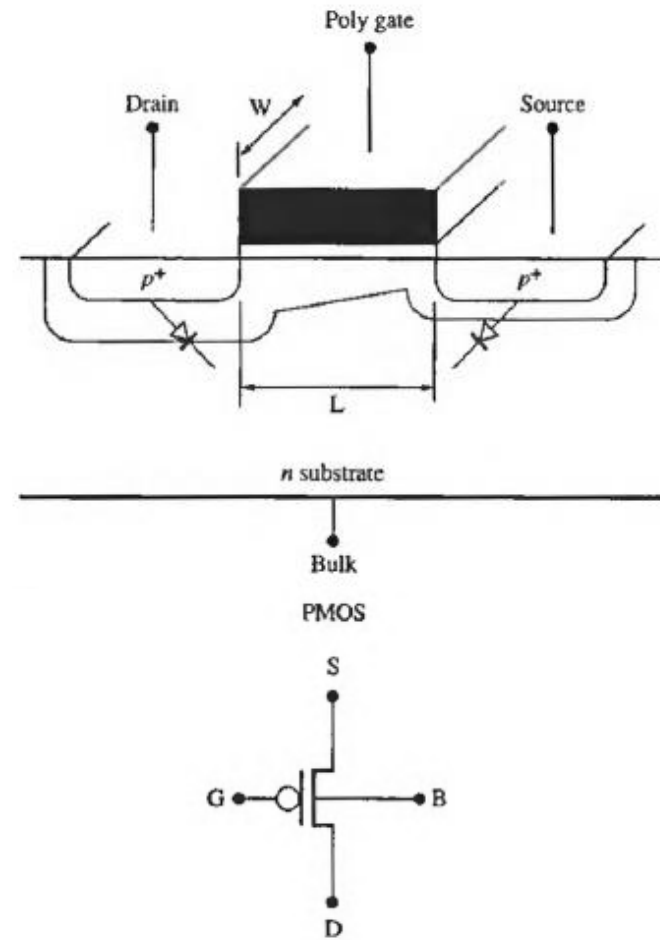
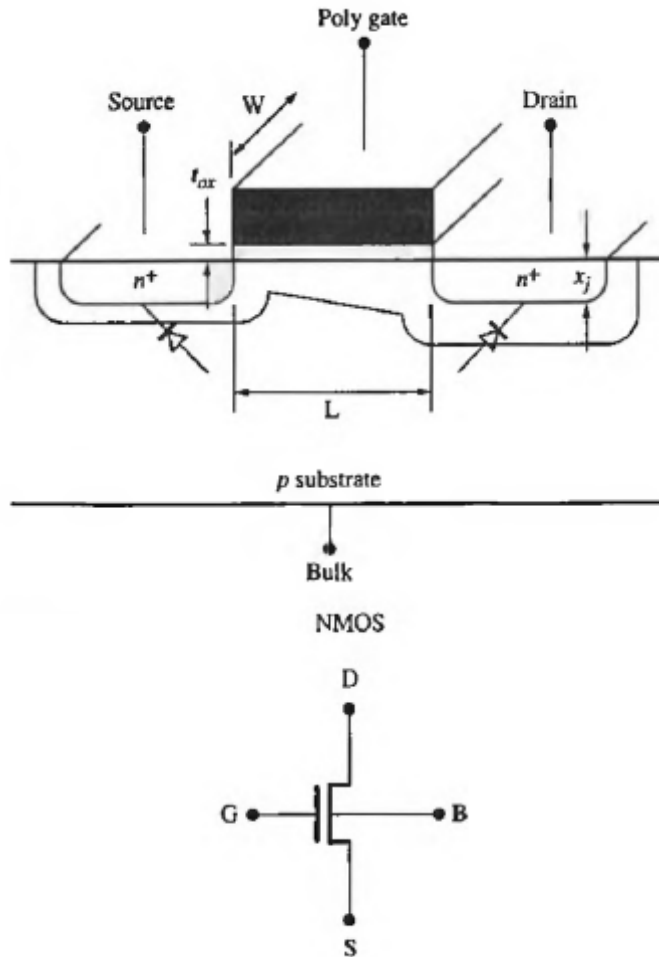
Current Ratio
of 2 1X devices



$$\frac{I_{DsatN}}{I_{DsatP}} = \frac{W_N v_{sat} C_{ox} (V_{GS} - V_{TN})^2 / (V_{GS} - V_{TN} + E_{CN} L_N)}{W_P v_{sat} C_{ox} (V_{GS} - V_{TP})^2 / (V_{GS} - V_{TP} + E_{CP} L_P)}$$

$$\frac{I_{DsatN}}{I_{DsatP}} = 2.4$$

NMOSFET and PMOSFET



DSM MOSFET Current Summary

- MOSFET reaches saturation when channel is pinched off at drain end or carriers achieve velocity saturation
 - if $V_{DS} \geq \frac{(V_{GS} - V_T) E_c L}{(V_{GS} - V_T) + E_c L} \Rightarrow \text{saturation}$
 - if $V_{DS} < \frac{(V_{GS} - V_T) E_c L}{(V_{GS} - V_T) + E_c L} \Rightarrow \text{linear}$

$$I_{DS} = W v_{sat} C_{ox} \frac{(V_{GS} - V_T)^2}{(V_{GS} - V_T) + E_c L} \quad \text{saturation}$$

Including Channel Length Modulation,

$$I_{DS} = W v_{sat} C_{ox} \frac{(V_{GS} - V_T)^2}{(V_{GS} - V_T) + E_c L} (1 + \lambda V_{DS}) \quad \text{saturation}$$

$$I_{DS} = \frac{W}{L} \cdot \frac{\mu_c C_{ox}}{\left(1 + \frac{V_{DS}}{E_c L}\right)} \left(V_{GS} - V_T - \frac{V_{DS}}{2}\right) V_{DS} \quad \text{linear}$$

- Reduce back to long-channel eqs. when $E_c L \gg V_{GS}, V_{DS}$

DSM MOSFET Current Summary

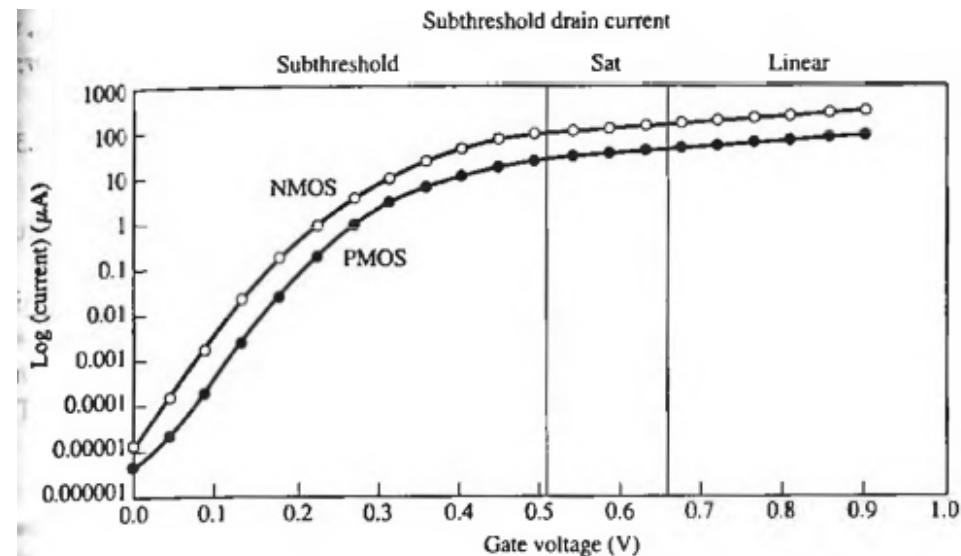
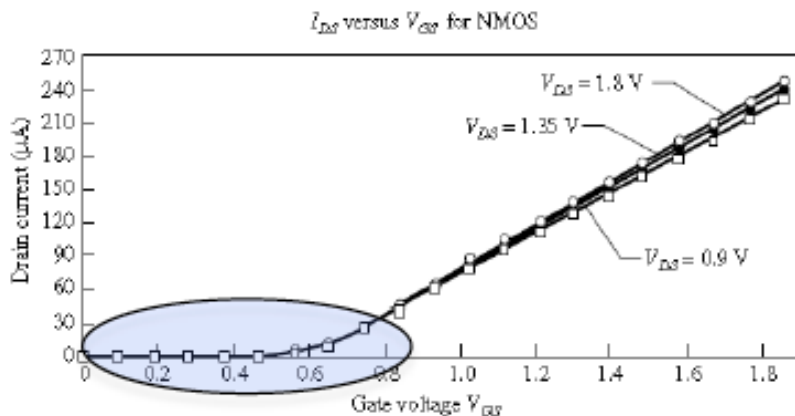
$$I_{DS}(\text{linear}) = \mu C_{ox} \frac{W}{L} \left[(V_{gs} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right] \times \frac{1}{1 + \frac{V_{DS}}{E_c L}} \quad \text{as } E_c L \rightarrow \infty, \frac{1}{1 + \frac{V_{DS}}{E_c L}} \nearrow 1$$

$$I_{DS}(\text{sat}) = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{gs} - V_T)^2 \times \frac{E_c L}{V_{gs} - V_T + E_c L} \quad \text{as } E_c L \rightarrow \infty, \frac{E_c L}{V_{gs} - V_T + E_c L} \nearrow 1$$

$\lim_{E_c L \rightarrow \infty} I_{DS} = I_{DS}$ long channel equations

Subthreshold Conduction

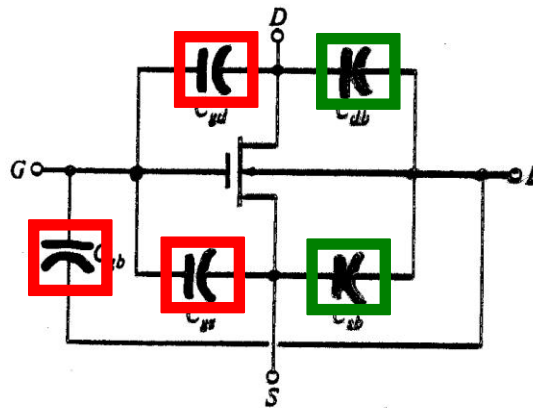
- MOSFET turns on gradually. At or below threshold, there is significant drain leakage current.



$$I_{sub} = I_s e^{\frac{q(V_{GS} - V_T - V_{eff})}{nkT}} \left(1 - e^{\frac{-qV_{DS}}{kT}} \right) \rightarrow S = \Delta V_{GS} = \frac{nkT}{q} \ln(10)$$

- What is the ideal n and S ?

MOSFET Capacitances



 oxide cap.
 jn. cap.

FIGURE 2.6
 MOS transistor capacitances. $C_g = C_{gs} + C_{gb} + C_{gd}$.

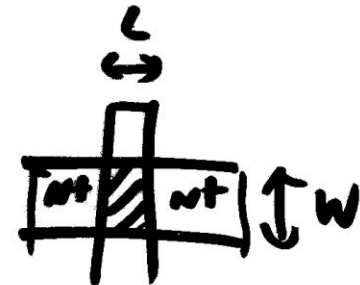
$C_{sb}, C_{db} \sim n^+p$ junction capacitances

$$C_g = C_{gs} + C_{gb} + C_{gd}$$

$$\approx WL C_{ox} = WL \epsilon_{ox} / t_{ox}$$

$$\propto \frac{1}{t_{ox}}$$

$$\propto W \text{ or } L$$



MOSFET Capacitances

Gate Capacitance

$$C_G = WLC_{ox} = WL(\epsilon_{ox}/t_{ox})C_{ox} \\ = WC_g$$

For 5 μ m tech ($t_{ox} = 110$ nm):

$$C_g \approx 1.6\text{fF}/\mu\text{m}$$

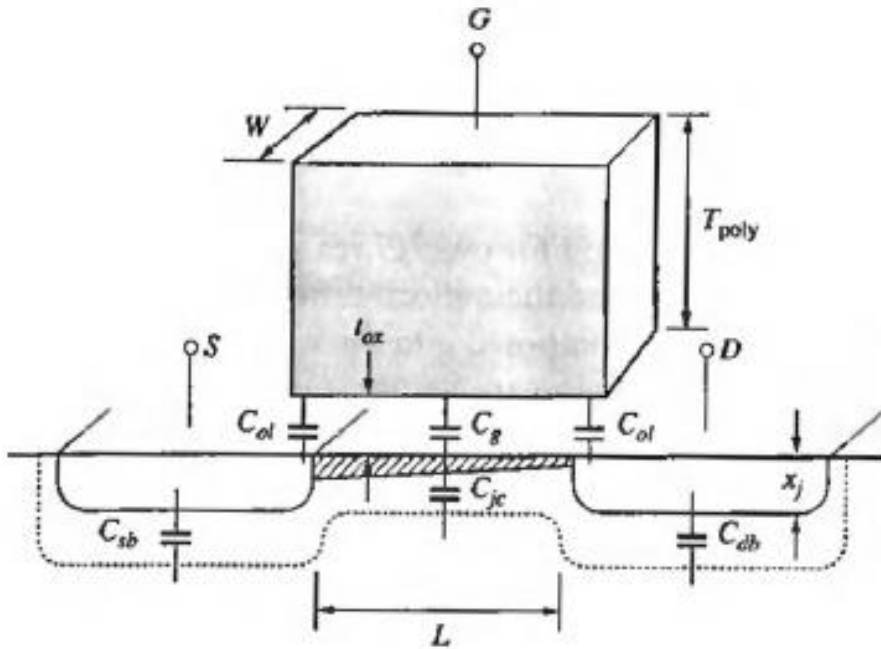
For 350nm tech ($t_{ox} = 7.5$ nm):

$$C_g \approx 1.6\text{fF}/\mu\text{m}$$

For 130nm tech ($t_{ox} = 2.2$ nm):

$$C_g \approx 1.6\text{fF}/\mu\text{m}!$$

C_g remains constant is a consequence of MOSFET scaling!
Factor L/t_{ox} remains constant



Gate Capacitances

Thin-Oxide Gate Capacitance

$t_{ox} = 2.2\text{nm}$, $W = 400\text{nm}$,

$L = 100\text{nm}$

Total: $C_G = WC_g$
 $= 1.6\text{fF}/\mu\text{m} \times 0.4 \mu\text{m} = 0.64\text{fF}$

Cutoff:

$C_{GS} = 0$, $C_{GD} = 0$

$C_{GB} \approx 0.64\text{fF}/2 = 0.32\text{fF}$

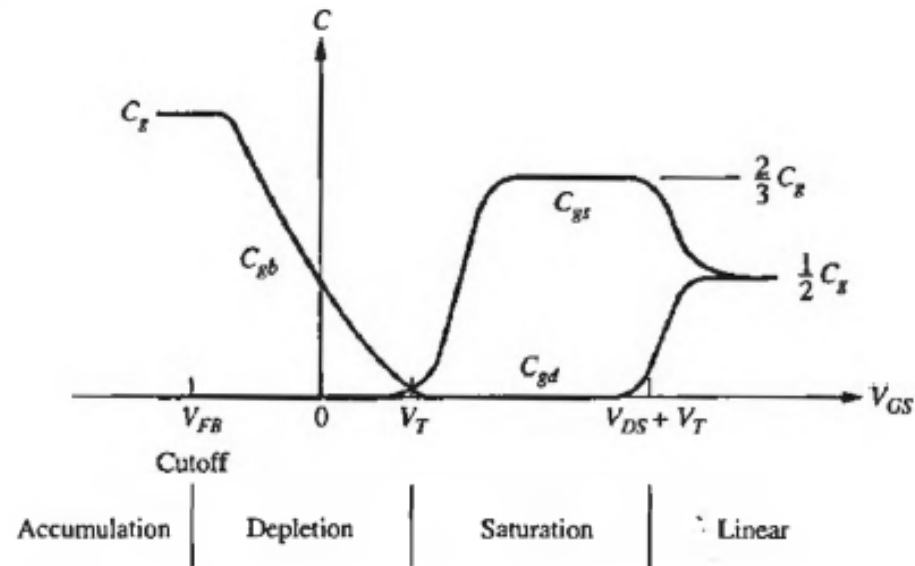
Linear:

$C_{GS} = 0.32\text{fF}$, $C_{GD} = 0.32\text{fF}$

$C_{GB} = 0$

Saturation:

$C_{GS} = 0.43\text{fF}$, $C_{GD} = 0$, $C_{GB} = 0$



Junction Capacitances

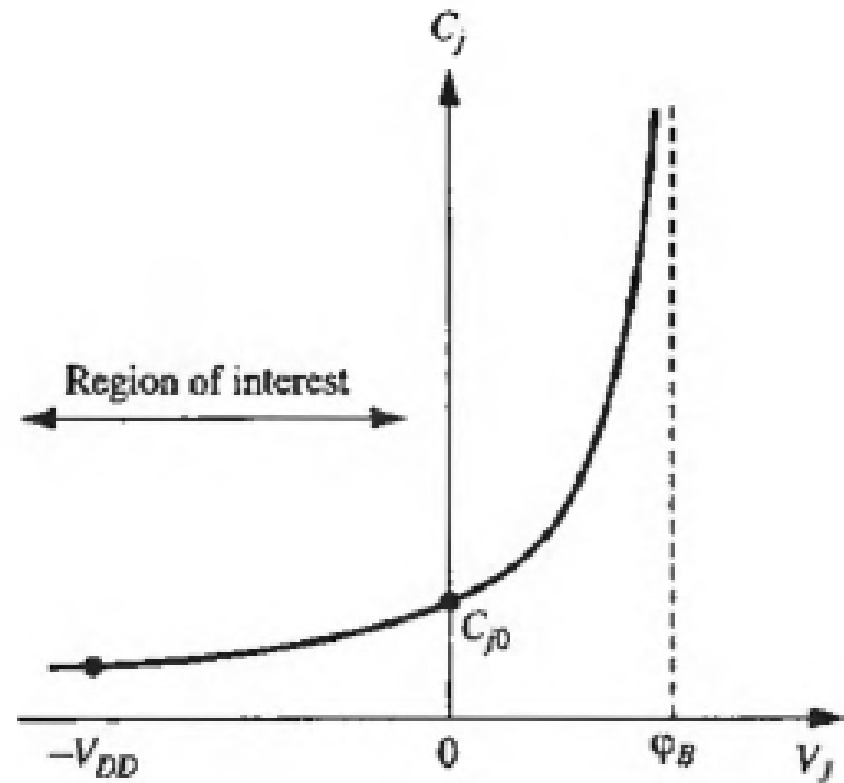
Built-in Junction Potential

$$\phi_B = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2}$$

Junction Capacitance

$$C_J = \frac{C_{j0} A}{\left(1 - \frac{V_J}{\phi_B}\right)^m}$$

$$C_{j0} = \sqrt{\frac{\epsilon_{si} q}{2\phi_B} \frac{N_A N_D}{N_A + N_D}} \approx \sqrt{\frac{\epsilon_{si} q N_A}{2\phi_B}}$$



Junction Capacitances

Junction Area Estimates:

Junction Bottom

$$A_b = W Y$$

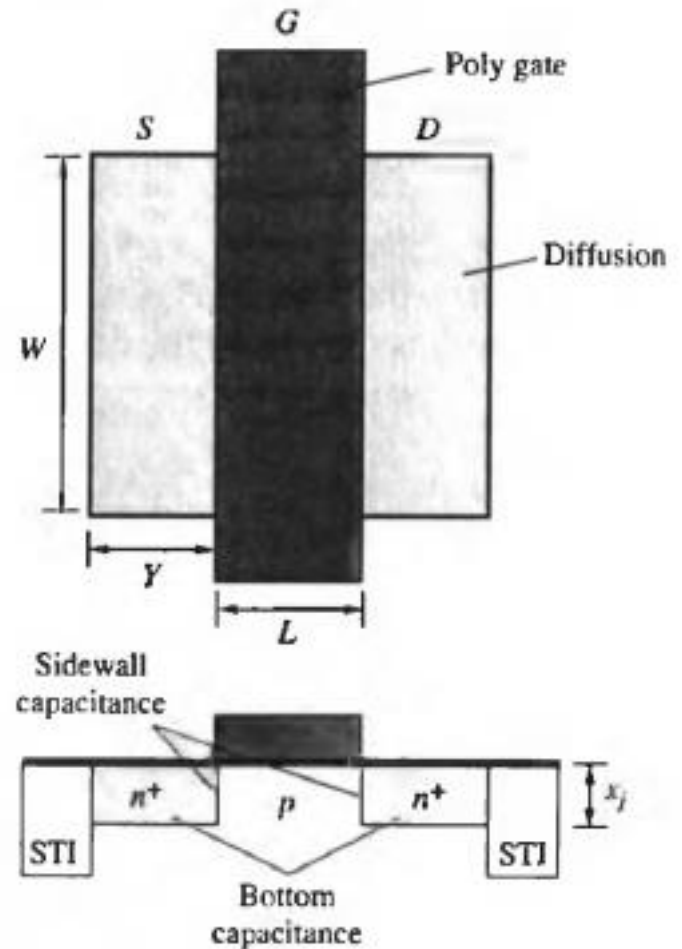
Junction Sidewall

$$A_{sw} = W x_j$$

Total

$$C_J = \frac{C_{jb} A_b}{\left(1 - \frac{V_J}{\phi_{Bb}}\right)^{mj}} + \frac{C_{jsw} A_{sw}}{\left(1 - \frac{V_J}{\phi_{Bsw}}\right)^{mjsw}}$$

$$C_J = \frac{C_{jb} A_b + C_{jsw} A_{sw}}{\left(1 - \frac{V_J}{\phi_B}\right)^{mj}} = \frac{C_{jb} (A_b + A_{sw})}{\left(1 - \frac{V_J}{\phi_B}\right)^{mj}}$$



Junction Capacitances

$$C_{eq} = \frac{Q_j(V_2) - Q_j(V_1)}{V_2 - V_1} = \frac{\Delta Q}{\Delta V}$$

where

$$\Delta Q = \int_{V_1}^{V_2} C(V) dV = \int_{V_1}^{V_2} C_{jb} \left(1 - \frac{V}{\phi_B}\right)^{-m} dV$$

Thus

$$C_{eq} = - \frac{C_{jb} \phi_B}{(V_2 - V_1)(1 - m)} \left[\left(1 - \frac{V_2}{\phi_B}\right)^{1-m} - \left(1 - \frac{V_1}{\phi_B}\right)^{1-m} \right]$$

Good for hand calculations:

$$C_j = K_{eq} C_{jb} (Y + x_j) W = C_j W$$

We are now able to define K_{eq} , the dimensionless constant used to relate C_{eq} to C_{jb} for specified values of V_1 and V_2 . In the case of an abrupt junction, $m = 1/2$, and therefore

$$K_{eq} = \frac{C_{eq}}{C_{jb}} = \frac{-2\phi_B^{1/2}}{V_2 - V_1} [(\phi_B - V_2)^{1/2} - (\phi_B - V_1)^{1/2}] \quad (2.43)$$

Fortunately, this complicated equation is required in only one case. Once computed, the simplified equation for hand calculations is as follows:

$$C_j = K_{eq}(C_{jb} W Y + C_{jb} x_j W) = K_{eq}(C_{jb} Y + C_{jb} x_j) W = K_{eq} C_{jb} (Y + x_j) W$$

Junction Caps. (HJS Ex. 2.12)

Problem:

- (a) Find ϕ_B and C_{jb} for an n^+p junction diode with $N_D = 10^{20} \text{ cm}^{-3}$ and $N_A = (3)10^{17} \text{ cm}^{-3}$.

Solution:

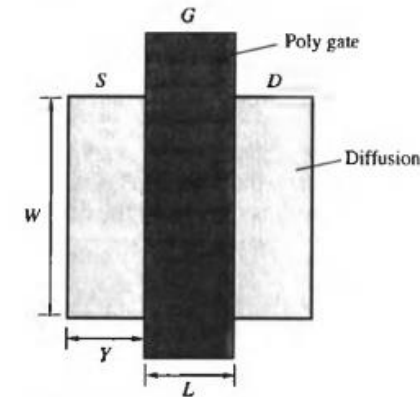
From Equation (2.37),

$$\phi_B = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2} = 0.026 \ln \left(\frac{3(10^{17})(10^{20})}{(1.45(10^{10}))^2} \right) = 1 \text{ V}$$

From Equation (2.39)

$$\begin{aligned} C_{jb} &= \sqrt{\frac{\epsilon_{si} q}{2\phi_B} \frac{N_A N_D}{N_A + N_D}} \approx \sqrt{\frac{\epsilon_{si} q N_A}{2\phi_B}} \\ &= \sqrt{\frac{11.7 * (8.85)(10^{-14}) * 1.6(10^{-19})(3)(10^{17})}{2(1.0)}} \approx 1.6 \frac{\text{fF}}{\mu\text{m}^2} \end{aligned}$$

Junction Caps. (HJS Ex. 2.12)



Problem:

- (b) For a $0.13\ \mu\text{m}$ process, $W = 400\ \text{nm}$, $L = 100\ \text{nm}$, $x_j = 50\ \text{nm}$, and the diffusion extension is $Y = 300\ \text{nm}$. Using the layout of Figure 2.20, find C_j in units of fF for $V_j = 0$ and $V_j = -1.2\ \text{V}$.

Solution:

For $V_j = 0$, the value is obtained by multiplying C_{jb} with $(Y + x_j)W$

$$C_j = C_{jb}(Y + x_j)W = 1.6\ \text{fF}/\mu\text{m}^2 \times (0.3\ \mu\text{m} + 0.05\ \mu\text{m}) \times 0.4\ \mu\text{m} \approx 0.22\ \text{fF}$$

For $V_j = -1.2$,

$$\begin{aligned} C_j &= \frac{C_{jb}(Y + x_j)W}{(1 - V_j/\phi_B)^m} \\ &= \frac{1.6\ \text{fF}/\mu\text{m} \times (0.3\ \mu\text{m} + 0.05\ \mu\text{m}) \times 0.4\ \mu\text{m}}{(1 + 1.2/1.0)^{1/2}} = 0.16\ \text{fF} \end{aligned}$$

Junction Caps. (HJS Ex. 2.12)

Problem:

(c) Find K_{eq} for $V_1 = -1.2\text{ V}$, $V_2 = 0$ and then compute the large-signal effective C_j .

Solution:

From Equation (2.43),

$$K_{eq} = \frac{-2(1)^{1/2}}{0 - (-1.2)} [(1 - 0)^{1/2} - (1 - (-1.2))^{1/2}] = 0.8$$

To compute C_j , we use

$$\begin{aligned}\therefore C_j &= K_{eq} C_{j0} (V + \phi_j) W \\ &= 0.8 \times 1.6 \text{ fF}/\mu\text{m}^2 \times (0.3\mu\text{m} + 0.05\mu\text{m}) \times 0.4\mu\text{m} \approx 0.18 \text{ fF}\end{aligned}$$

Overlap Capacitance

Total: $C_{ol} = C_{ov} + C_f$
(Overlap and fringing caps.)

Overlap Capacitance

$$C_{ov} = C_{ox} \times L_D$$

$$C_f = \frac{2\epsilon_{ox}}{\pi} \ln \left(1 + \frac{T_{poly}}{t_{ox}} \right)$$

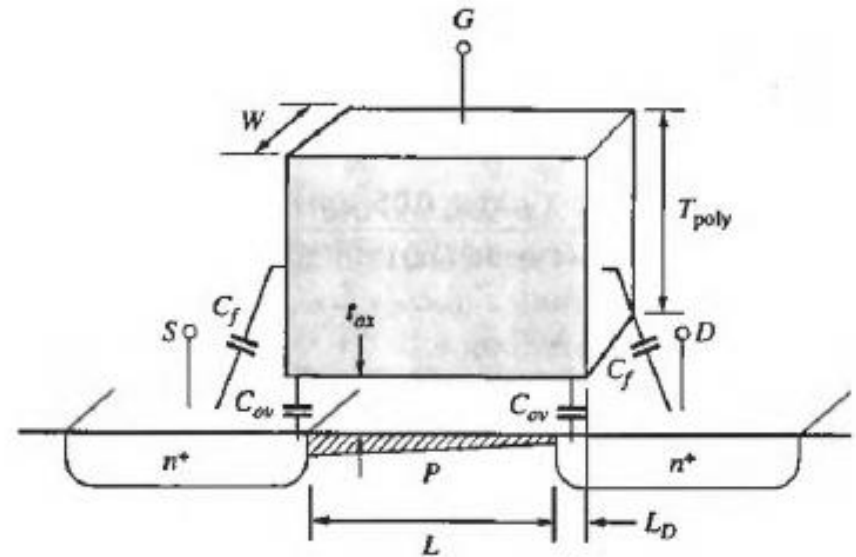
Ex. 2.13 (130nm tech.)

$$T_{poly}/t_{ox} = 100, L_D = 10\text{nm}$$

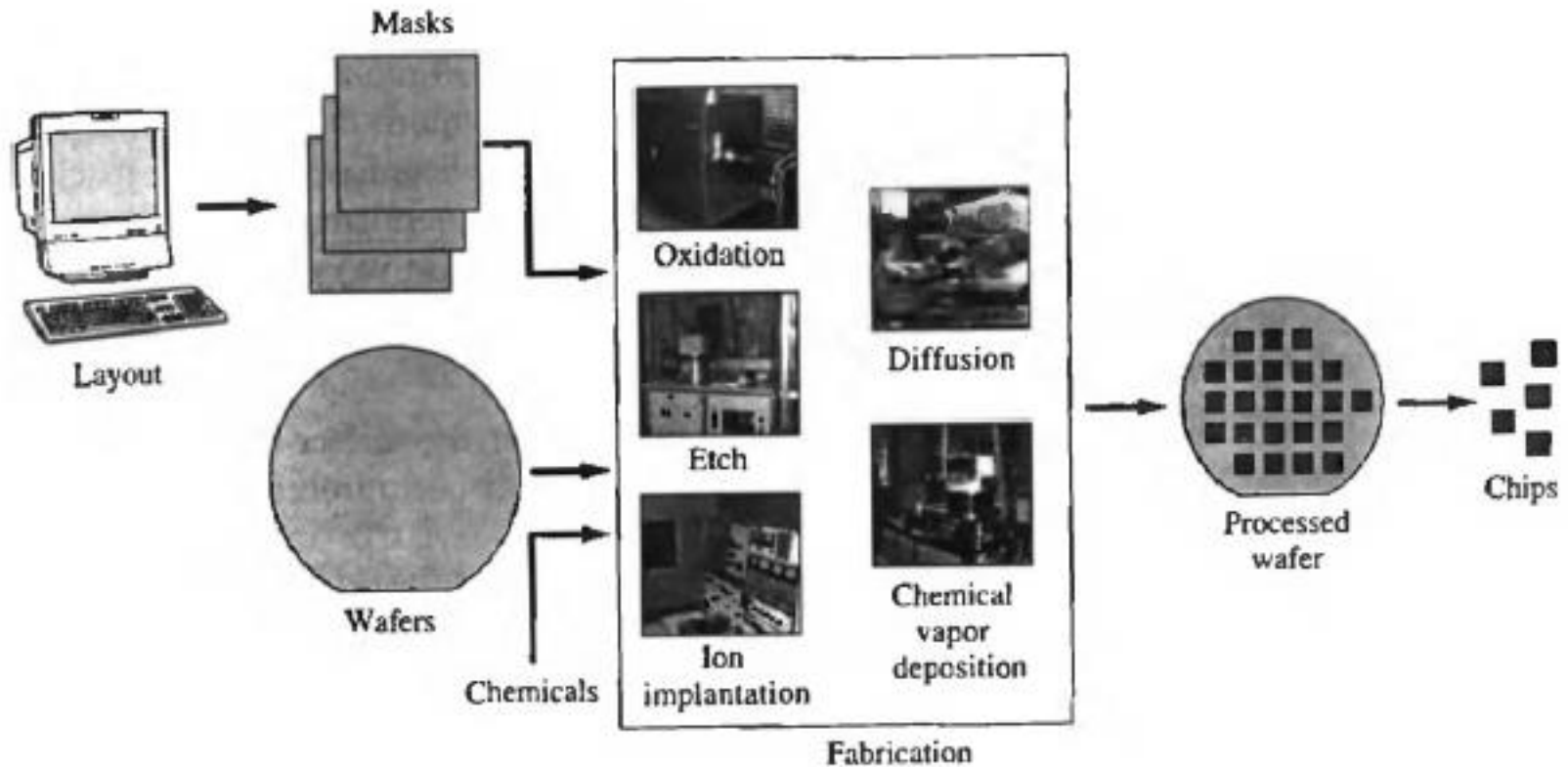
$$C_f = \frac{2(4)(8.85 \times 10^{-14})}{3.14} \ln(1 + 100) \approx 0.1 \text{ fF}/\mu\text{m}$$

$$C_{ov} = C_{ox} L_D \approx 0.15 \text{ fF}/\mu\text{m}.$$

$$C_{ol} = 0.1 \text{ fF}/\mu\text{m} + 0.15 \text{ fF}/\mu\text{m} = 0.25 \text{ fF}/\mu\text{m}$$



IC Fabrication Process



CMOSFET Structure

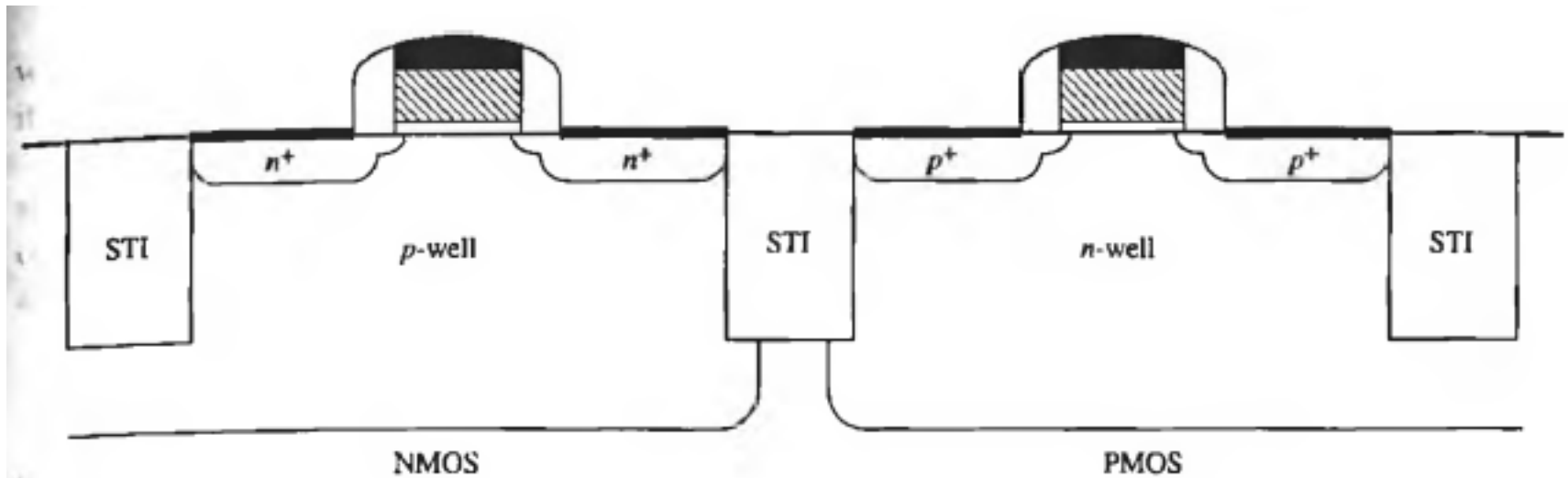
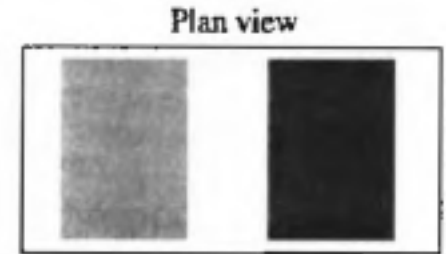
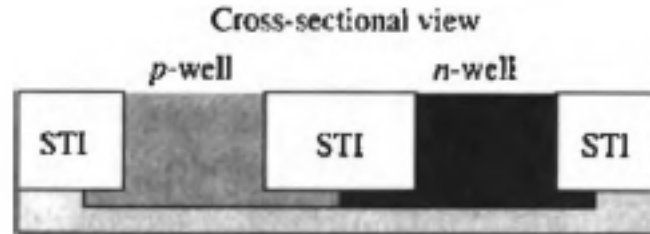


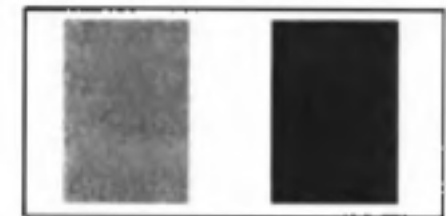
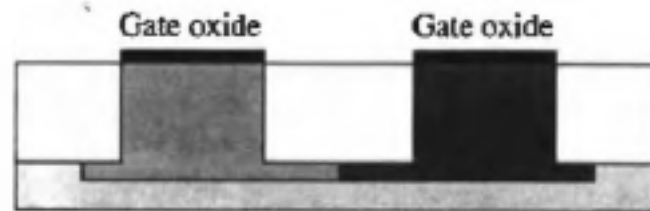
Figure 2.3
CMOS transistors structure.

IC Fabrication Process (Frontend FEOL)

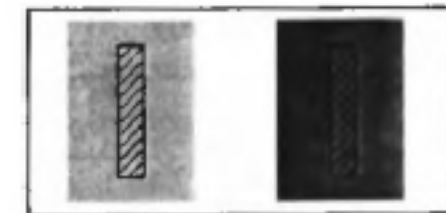
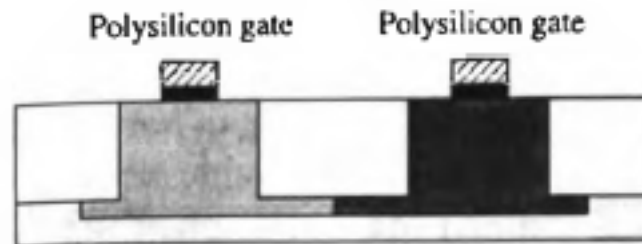
1. Well Formation,
Field Oxide
Isolation



2. Gate Oxidation

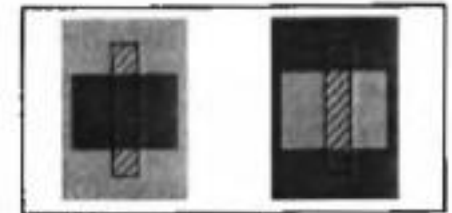
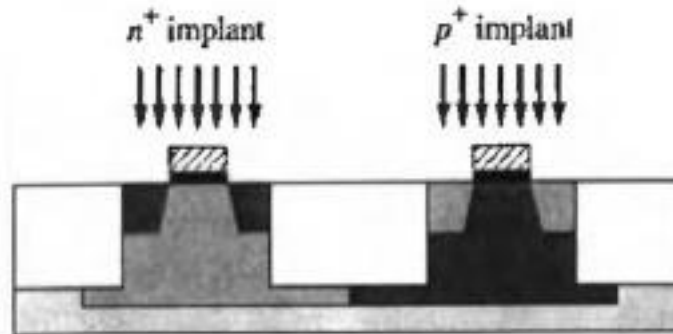


3. Poly-Si Gate
Definition

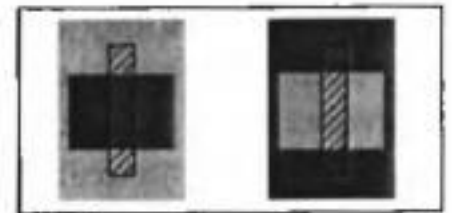
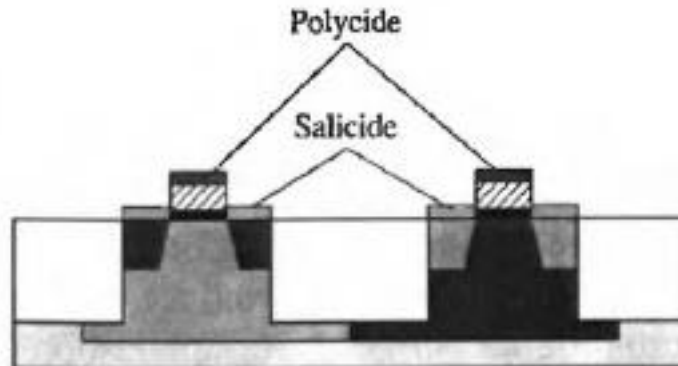


IC Fabrication Process (Frontend FEOL)

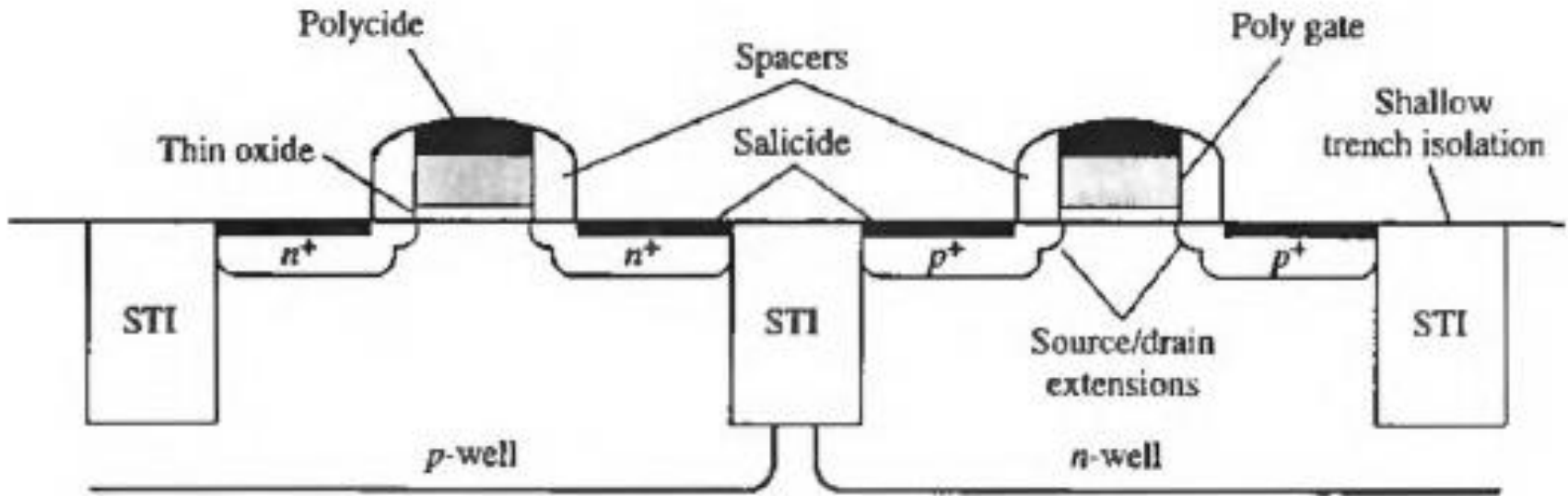
4. Source/Drain Implantation



5. Gate and S/D Silicidation

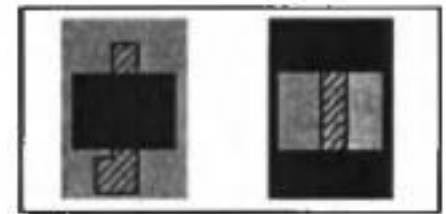
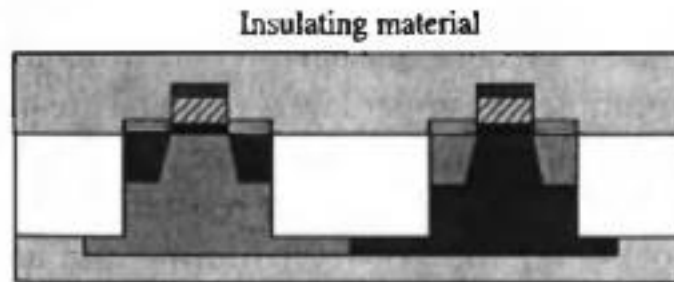


DSM CMOS Transistor Structures

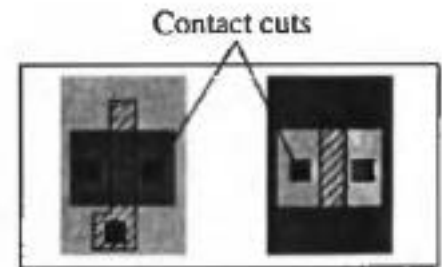
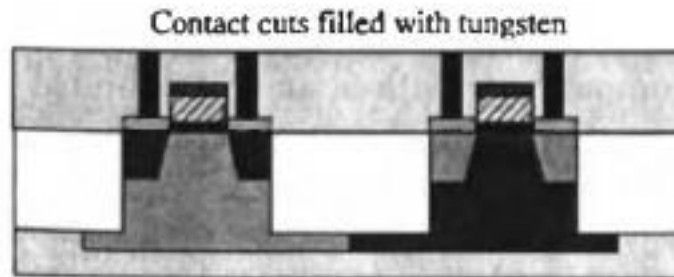


IC Fabrication Process (Backend BEOL)

1. Dielectric Planarization

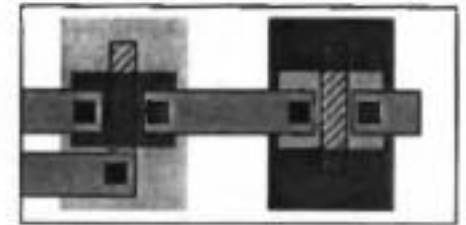
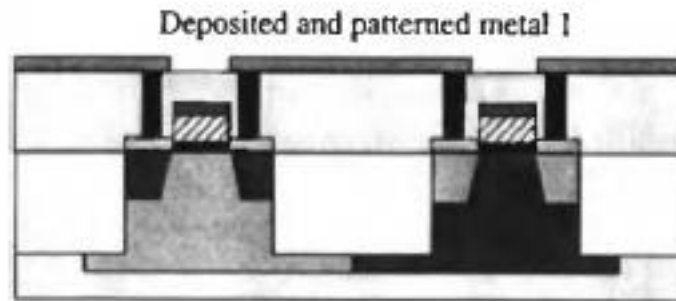


2. Tungsten Plug Formation

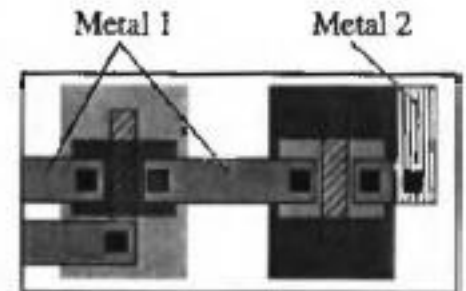
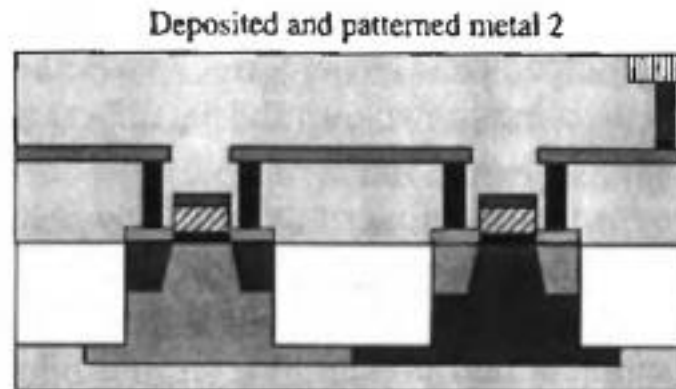


IC Fabrication Process (Backend BEOL)

3. Metal 1
Deposition and
Patterning

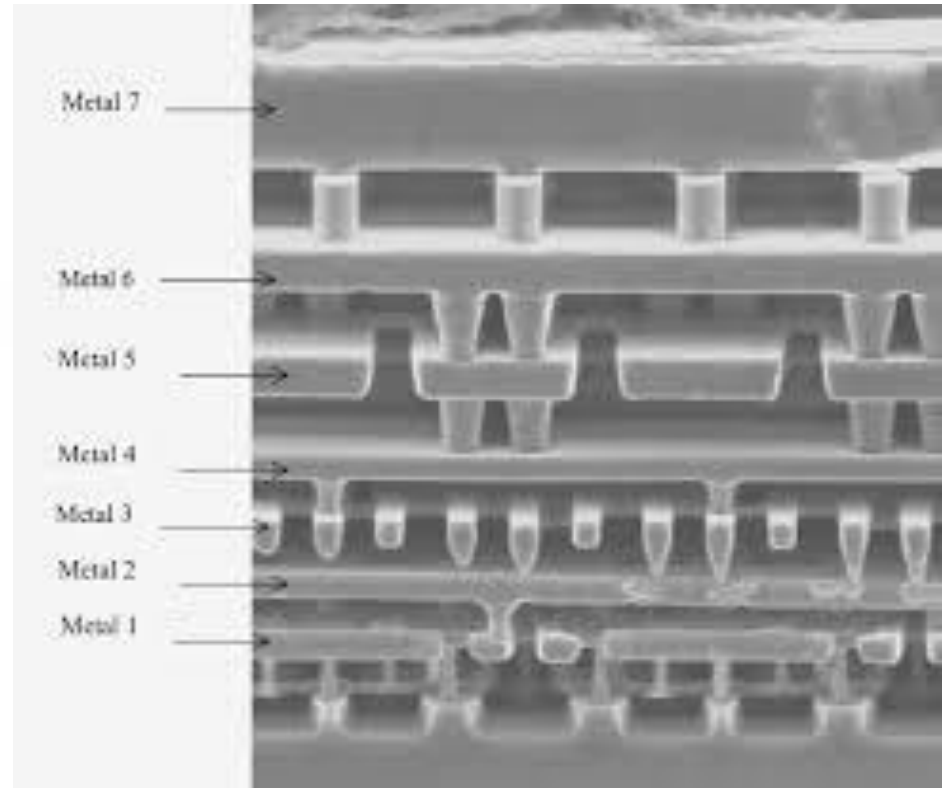
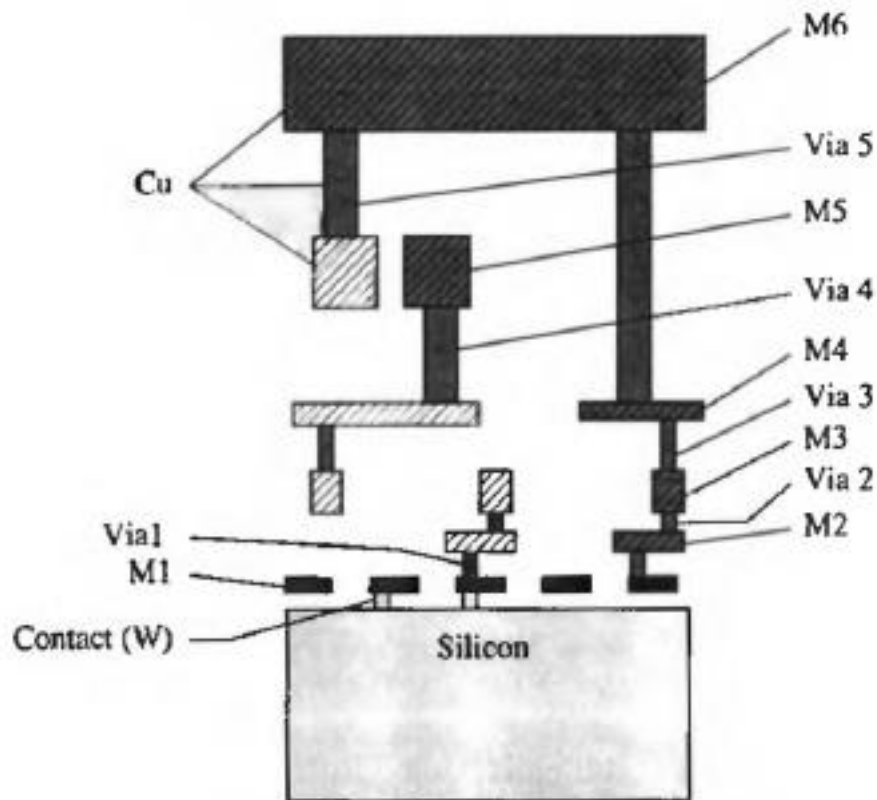


4. Metal 2
Deposition and
Patterning



5. Metal 3...

Multilevel Metal Interconnect



Multilevel Metal Interconnect

